



Yiheng Lyu

Passport: EN4764545 | **Date of birth:** 1 Jun 2004 | **Place of birth:** Shanghai, China |

Nationality: Chinese | **Gender:** Male | **Phone number:** (+86) 18918557683 (Home) |

Email address: xheng040601@outlook.com | **Address:** Shanghai, China (Home)

EDUCATION AND TRAINING

3 SEP 2022 – CURRENT Nanjing, China

B.ENG. Nanjing Tech Univeristy

Website <https://www.njtech.edu.cn/> | **Field of study** Electronics and automation | **Final grade** 88/100 |

Level in EQF EQF level 6 | **Thesis** Adaptive control of multi-joint robotic manipulator systems with RBF neural networks

RESEARCH EXPERIENCE

OCT 2024 – NOV 2025

Lyapunov Stability Analysis of Distributed Adaptive Resource Allocation under Lipschitz Continuous Costs via Saddle-Point Dynamics

- Extended the distributed adaptive resource allocation framework from Simone Baldi, Southeast University by relaxing the strong convexity assumption on $\nabla f(x)$ to Lipschitz continuity, enabling convergence analysis under weaker conditions.
- Designed a cross-term Lyapunov function and leveraged pseudoinverse-based analysis to rigorously prove stability of the saddle-point dynamics in constrained multi-agent systems.
- Conducted Python simulations to validate theoretical results and demonstrate convergence performance under realistic system parameters.
- Applied the theoretical framework to multi-joint robotic manipulator control, bridging adaptive resource allocation research with practical cyber-physical system applications.

MAR 2025 – NOV 2025

Power Flow Scheduling and Load Balancing in Electric Vehicle Charging Stations

- Developed a hierarchical EV-grid scheduling framework for smart charging and reactive power optimization
- Simulated EV SOC, OD matrices, and charging start-time distributions; designed upper-lower layer coordination mechanism
- Implemented distributed projection gradient method (DPGM) and finite-time consensus algorithm (FACA)
- Verified voltage stability, load balancing, and renewable utilization via Monte Carlo scenario reduction and IEEE node simulation

WORK EXPERIENCE

AUTOMATION ENGINEER – 3M SOLVENTUM – 22 Jul 2024 – 31 Aug 2024 – SHANGHAI, CHINA

Address: Pu Jiang Plant, 35# Xin Jun Huan Road | Shang Hai, PR China, CHN 3M Medical Devices and Materials Mfg (SH) Co., Ltd, Shanghai, China

- Completed Rockwell RSLogix 5000 PLC training, acquiring skills in industrial automation programming and ladder logic development.
- Contributed to the paper quality inspection system project by using industrial cameras for image acquisition and applying machine learning algorithms to detect black spots/defects and crop affected areas.
- Assisted in maintenance and optimization of motion control and process control systems, including robotic arms, pneumatic conveyors, tension regulation, and radiation-based thickness measurement automation.
- Learned and applied Cisco Converged Plantwide Ethernet (CPwE) industrial switches and PCN (Process Control Network) architecture for reliable plant-floor networking.

- Gained hands-on experience operating SCADA systems for real-time monitoring and control of industrial processes.

● **PROJECTS**

JUL 2024 – SEP 2024

PLC Industrial Automation - 3M Paper Quality Inspection System

Implemented real-time defect detection with industrial cameras and PLC control (Rockwell RSLogix 5000). Optimized paper cutting position control and coordinated pneumatic and robotic actuators.

JUN 2025 – JUL 2025

PLC Industrial Automation - Industrial Gas Detection and Auto-Blowback System

Designed CO concentration monitoring system with Siemens S7-1200 PLC and HMI; implemented adjustable closed-loop pump control, alarm response mechanism, and manual/automatic operation modes.

SEP 2025 – OCT 2025

PLC Industrial Automation - Industrial Mixing and Feeding Automation

Designed ladder logic and WinCC HMI for automated material feeding, mixing, and transfer; implemented manual/automatic switching and data logging.

SEP 2025 – OCT 2025

PLC Industrial Automation - Hydraulic Drive Lab Platform Automation

Developed PLC control for hydraulic actuator coordination; implemented sensor-based closed-loop control, automatic/manual/alarm modes, and WinCC visualization.

NOV 2023 – APR 2024

Microcontroller Hardware Development - Hand Recognition Autonomous Control (Mediapipe + STM32)

Used Python Mediapipe for hand keypoint detection; mapped gestures to control signals. Designed STM32 hardware and control program, achieving <50ms low-latency response.

NOV 2024 – DEC 2024

Microcontroller Hardware Development - STM32 High-Speed Data Acquisition Instrument

Implemented ADC + DMA signal acquisition and PWM output; supported continuous high-frequency data collection (>1h) with <0.1% packet loss; visualized via Sofa software.

JUN 2024 – JUN 2024

Microcontroller Hardware Development - 51 Electronic Board Ultrasonic Rangefinder

Write here the description...Designed PCB and schematic in AD, Multisim, and Simulink; programmed in Keil for distance measurement. Achieved ±5cm accuracy and <100ms response time.

MAY 2025 – JUL 2025

Power Electronics & Electrical Control - Photovoltaic System Design via Street Image Recognition

Used Python to analyze rooftop areas for solar irradiance; optimized PV array layout. Simulated multi-stage inverter topologies in MATLAB-Simulink; designed automatic power dispatch strategies.

● **LANGUAGE SKILLS**

Mother tongue(s): **CHINESE**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● **CERTIFICATIONS**

British Council; IDP IELTS; and Cambridge University Press & Assessment
IELTS Band 7.0

COMAP – The Consortium for Mathematics and its Applications, USA
MCM - Honorable Mention

Jiangsu Provincial Education Examination Authority, China
NCRE Network Technology Level 3

China Association of Higher Education
RAICOM Robotics Competition – Provincial Second Prize

Computer Education Research Association of Chinese Universities
A-ST Network Technology Competition – Provincial Third Prize
